

GAZ - Temă seminar 9

1. $Q: \mathbb{R}^3 \rightarrow \mathbb{R}$

$$Q(x) = x_1^2 + 3x_2^2 + x_3^2 - 2x_1x_2 - 4x_2x_3 \quad (\forall) x = (x_1, x_2, x_3) \in \mathbb{R}^3$$

a) $A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 1 \end{pmatrix} \rightarrow$ matrice asociată pătrii a în raport cu baza canonică

$$\begin{aligned} Q(x) &= (x_1^2 - 2x_1x_2 + x_2^2) + (4x_2^2 - 4x_2x_3 + x_3^2) - 2x_2^2 \\ &= (x_1 - x_2)^2 + (2x_2 - x_3)^2 - 2x_2^2 \end{aligned}$$

efectul schimbării de coordonate

$$\begin{cases} y_1 = x_1 - x_2 \\ y_2 = 2x_2 - x_3 \\ y_3 = x_2 \end{cases}$$

$\Rightarrow Q(x) = y_1^2 + y_2^2 - 2y_3^2, \quad (\forall) x = (y_1, y_2, y_3) \in \mathbb{R}^3$

$\downarrow$
coordonate în raport cu noua bază de raportare

b) $A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 1 \end{pmatrix} \rightarrow$ m. asociată lui Q_1

$\rightarrow \Delta_1 = 1$

$\Delta_2 = \begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix} = 2$

$\Delta_3 = \det A = \begin{vmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 1 \end{vmatrix} = -2$

$$\Rightarrow Q(x) = \frac{1}{\Delta_1} (x_1')^2 + \frac{\Delta_1}{\Delta_2} (x_2')^2 + \frac{\Delta_2}{\Delta_3} (x_3')^2$$

$$= \frac{1}{1} (x_1')^2 + \frac{1}{2} (x_2')^2 + 1 (x_3')^2, (\forall) x = (x_1', x_2', x_3') \in \mathbb{R}^3$$

$$2. \quad A = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \end{pmatrix} \quad \Rightarrow \quad C^t = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 2 \\ 1 & 3 & -3 \end{pmatrix}$$

$$A' = C^T A C$$

$$= \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 2 \\ 1 & 3 & -3 \end{pmatrix} \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -3 & -2 \\ 6 & -9 & 2 \\ -4 & 7 & -6 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} -5 & -1 & -3 \\ -1 & 25 & -27 \\ -3 & -27 & 35 \end{pmatrix} \quad \rightarrow \text{in axes of Eulerian system in rapport in noua baza}$$

3. fie $F: \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$

$$F(x, y) = 2x_1y_1 + x_2y_2 - 2x_1y_2 - 2x_2y_1 - 2x_2y_3 - 2x_3y_2$$

a) $x = (x_1, x_2, x_3) \in \mathbb{R}^3$

$y = (y_1, y_2, y_3) \in \mathbb{R}^3$

a, b) dem linear lui F in ambele arg:

fie $x, y, z \in \mathbb{R}^3$, $x = (x_1, x_2, x_3)$

in $\alpha, \beta \in \mathbb{R}$ $y = (y_1, y_2, y_3)$

$z = (z_1, z_2, z_3)$

$$\begin{aligned} \text{ii)} \quad F(\alpha x + \beta y, z) &= 2(\alpha x_1 + \beta y_1)z_1 + (\alpha x_2 + \beta y_2)z_2 - 2(\alpha x_1 + \beta y_1)z_2 - \\ &- 2(\alpha x_2 + \beta y_2)z_1 - 2(\alpha x_2 + \beta y_2)z_3 - 2(\alpha x_3 + \beta y_3)z_2 = \\ &= \alpha(2x_1z_1 + x_2z_2 - 2x_1z_2 - 2x_2z_1 - 2x_2z_3 - 2x_3z_2) + \\ &+ \beta(2y_1z_1 + y_2z_2 - 2y_1z_2 - 2y_2z_1 - 2y_2z_3 - 2y_3z_2) \\ &= \alpha F(x, z) + \beta F(y, z) \end{aligned}$$

$$\begin{aligned} \text{iii)} \quad F(x, \alpha y + \beta z) &= 2x_1(\alpha y_1 + \beta z_1) + x_2(\alpha y_2 + \beta z_2) - 2x_1(\alpha y_2 + \beta z_2) - \\ &- 2x_2(\alpha y_1 + \beta z_1) - 2x_2(\alpha y_3 + \beta z_3) - 2x_3(\alpha y_2 + \beta z_2) = \\ &= \alpha(2x_1y_1 + x_2y_2 - 2x_1y_2 - 2x_2y_1 - 2x_2y_3 - 2x_3y_2) + \\ &+ \beta(2x_1z_1 + x_2z_2 - 2x_1z_2 - 2x_2z_1 - 2x_2z_3 - 2x_3z_2) \\ &= \alpha F(x, y) + \beta F(x, z) \end{aligned}$$

$\Rightarrow$ F liniară în ambele arg

$A = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix}$ - matricea lui F în raport cu baza canonică

$A \rightarrow$ matrice sim (${}^t A = A$) $\Rightarrow F$ formă bil sim

$$c) B_1 = \{ (1, 1, 1), (2, -1, 2), (1, 3, -3) \}$$

$$C = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \end{pmatrix} \Rightarrow {}^t C = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 2 \\ 1 & 3 & -3 \end{pmatrix}$$

matrice bil sim in rep cu baza B_1 : $A' = {}^t C A C$

$$A' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 2 \\ 1 & 3 & -3 \end{pmatrix} \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -3 & -2 \\ 6 & -9 & 2 \\ -4 & 7 & -6 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} -5 & -1 & -3 \\ -1 & 25 & -27 \\ 3 & -27 & 35 \end{pmatrix} \rightarrow \text{m arec' bil sim in rep cu noua bază}$$

d) $Q: \mathbb{R}^3 \rightarrow \mathbb{R}$ prin met dedublării :

$$Q(x) = 2x_1^2 + x_2^2 - 4x_1x_2 - 4x_2x_3 \quad (x) = (x_1, x_2, x_3) \in \mathbb{R}^3$$

i) formă canonică pătratică - Gauss :

$$\begin{aligned} Q(x) &= \cancel{(4x_1^2 - 4x_1x_2 + x_2^2)} + (4x_2^2 - 4x_2x_3 + x_3^2) - \cancel{2x_1^2 - 4x_1^2 - x_3^2} \\ &= (2x_1 - x_2)^2 + (2x_2 - x_3)^2 - 2x_1^2 - 4x_1^2 - x_3^2 \\ &= (2x_1^2 - 4x_1x_2) + x_2^2 - 4x_2x_3 \\ &= (\sqrt{2}x_1 - \sqrt{2}x_2)^2 - x_2^2 - 4x_2x_3 \end{aligned}$$

$$Q(x) = (\sqrt{2}x_1 - \sqrt{2}x_2)^2 - (x_2^2 + 4x_2x_3) \\ = (\sqrt{2}x_1 - \sqrt{2}x_2)^2 - (x_2 + 2x_3)^2 + 4x_3^2$$

alec sch de coord $\left\{ \begin{array}{l} y_1 = \sqrt{2}x_1 - \sqrt{2}x_2 \\ y_2 = x_2 + 2x_3 \\ y_3 = x_3^2 \end{array} \right.$

$$\rightarrow Q(x) = y_1^2 - y_2^2 + 4y_3^2 \quad (\forall x = (y_1, y_2, y_3) \in \mathbb{R}^3)$$

word in $\mathbb{R}^3$ cu noua baze de raportare

forma canonică pătratică - Jordan

$$\left\{ \begin{array}{l} \lambda_1 = 2 \\ \lambda_2 = \begin{vmatrix} 2 & -2 \\ -2 & 1 \end{vmatrix} = -2 \\ \lambda_3 = -8 \end{array} \right. \quad \left| \lambda_i \neq 0, (\forall) i = \overline{1, 3} \right.$$

$$\Rightarrow Q(x) = \frac{1}{2} (x_1')^2 - (x_2')^2 + \frac{1}{4} (x_3')^2, \quad (\forall) x = (x_1', x_2', x_3')$$

word in $\mathbb{R}^3$ cu noua baze de raportare